

1.3 Matrix & Matrix Operations

$$\begin{array}{ccc} A & B & AB \\ m \times \delta & \delta \times n & = m \times n \end{array}$$

inside

outside

→ If we need column of AB

Full Matrix A and Desired col. B

→ If we need row of AB

Desired row of A and full matrix B

→ Note: Matrix product can be written in linear combinations.

→ Trace: Sum of main diagonal entries of matrix.

$$\rightarrow \text{Trace}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\rightarrow (A^t)^t = A$$

$$\rightarrow (AB)^t = B^t A^t$$

Example: 2 (Equality of Matrix)

Two matrix are supposed to be equal if they are of same size and their corresponding entries are same

$$A = \begin{bmatrix} 2 & 1 \\ 3 & x \end{bmatrix} \quad B = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & 0 \end{bmatrix}$$

- $\rightarrow A = B$ if $x = 5$
- $\rightarrow A \neq B$ if $x \in \mathbb{R} - \{5\}$
- $\rightarrow A \neq C$ since $2 \times 2 \neq 2 \times 3$
- $\rightarrow B \neq C$ since $2 \times 2 \neq 2 \times 3$

Example: 3 (Addition and Subtraction)

1. Addition:

Two matrix can be added only if they have same no of rows and columns.

2. Subtraction:-

same as addition.

$$A = \begin{bmatrix} 2 & 1 & 0 & 3 \\ -1 & 0 & 2 & 4 \\ 4 & -2 & 7 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -4 & 3 & 5 & 1 \\ 2 & 2 & 0 & -1 \\ 3 & 2 & -4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

$$A+B = \begin{bmatrix} -2 & 4 & 5 & 4 \\ 1 & 2 & 2 & 3 \\ 7 & 0 & 3 & 5 \end{bmatrix}, \quad A-B = \begin{bmatrix} 6 & -2 & -5 & 2 \\ -3 & -2 & 2 & 5 \\ 1 & -4 & 11 & -5 \end{bmatrix}$$

$\rightarrow A+C, A-C, B+C, B-C$ are undefined.

Example: 4 (Scalar Multiples)

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 3 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 & 7 \\ -1 & 3 & -5 \end{bmatrix}, C = \begin{bmatrix} 9 & -6 & 3 \\ 3 & 0 & 12 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & 6 & 8 \\ 2 & 6 & 2 \end{bmatrix}, (-1)B = \begin{bmatrix} 0 & -2 & -7 \\ 1 & -3 & 5 \end{bmatrix}, \frac{1}{3}C = \begin{bmatrix} 3 & -2 & 1 \\ 1 & 0 & 4 \end{bmatrix}$$

Example: 5 (Multiplying Matrices)

Two matrix can be multiplied only if

columns of matrix A = rows of matrix B

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \rightarrow 2 \times 3, B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix} \rightarrow 3 \times 4$$

$$AB = \begin{bmatrix} - & - & - & - \\ - & - & ? & - \end{bmatrix}$$

row 2 of A and col 3 of B

$$\begin{aligned} &= (2 \times 4) + (6 \times 3) + (0 \times 5) \\ &= 8 + 18 + 0 \\ &= 26 \end{aligned}$$

$\downarrow$
2x4

Example: 6

$$\begin{array}{ccc} A & B & C \\ 3 \times 4 & 4 \times 7 & 7 \times 3 \end{array}$$

- AB is defined with size of 3x7.
- BC is defined with size of 4x3.
- AC is not defined.
- CB is not defined.
- BA is not defined.

Example : 7 (Desired row Col of AB)

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

Find second column of AB

$$2^{\text{nd}} \text{ Col of AB} = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 7 \end{bmatrix}$$

$$2^{\text{nd}} \text{ Col of AB} = \begin{bmatrix} 1 - 2 + 28 \\ 2 - 6 + 0 \end{bmatrix}$$

$$2^{\text{nd}} \text{ Col of AB} = \begin{bmatrix} 27 \\ -4 \end{bmatrix}$$

Find 1st row of AB

$$1^{\text{st}} \text{ row of AB} = \begin{bmatrix} 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

$$1^{\text{st}} \text{ row of AB} = \begin{bmatrix} 4 + 0 + 8 & 1 - 2 + 28 & 4 + 6 + 20 & 3 + 2 + 8 \end{bmatrix}$$

$$1^{\text{st}} \text{ row of AB} = \begin{bmatrix} 12 & 27 & 30 & 13 \end{bmatrix}$$

Example : 8 (Matrix Product as linear Combination)

$$\begin{bmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

A X

$$2 \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -3 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ -3 \end{bmatrix}$$

Ax = Linear combinations of columns of A .

$$x \begin{bmatrix} \\ \\ \end{bmatrix} + y \begin{bmatrix} \\ \\ \end{bmatrix} + z \begin{bmatrix} \\ \\ \end{bmatrix}$$

1st col 2nd col 3rd col

Example: 9 (Cols of Product AB as a Linear combinations)

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{bmatrix}$$

1st Col of AB :-

$$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 0 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

2nd Col of AB :-

$$2^{\text{nd}} \text{ Col of } AB = \begin{bmatrix} 27 \\ -4 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 7 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

3rd Col of AB :-

$$3^{\text{rd}} \text{ Col of } AB = \begin{bmatrix} 30 \\ 26 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 5 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

4th col of AB :-

$$4^{\text{th}} \text{ col of AB} = \begin{bmatrix} 13 \\ 12 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + 2 \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Example: 10 (Column-Row Expansion)

Find the Column row expansion of the product

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \\ -3 & 5 & 1 \end{bmatrix} \rightarrow (1)$$

Columns of A :-

$$C_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, C_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

Rows of B:

$$r_1 = \begin{bmatrix} 2 & 0 & 4 \end{bmatrix}$$

$$r_2 = \begin{bmatrix} -3 & 5 & 1 \end{bmatrix}$$

Expansion Formula:-

$$\therefore AB = C_1 r_2 + C_2 r_1$$

Row Column Expansion:-

$$AB = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} \begin{bmatrix} -3 & 5 & 1 \end{bmatrix}$$

$2 \times 1 \quad 1 \times 3$

$$AB = \begin{bmatrix} 2 & 0 & 4 \\ 4 & 0 & 8 \end{bmatrix} + \begin{bmatrix} -9 & 15 & 3 \\ 3 & -5 & -1 \end{bmatrix} \rightarrow (2) \rightarrow \underline{\underline{\text{Ans}}}$$

Check eq (1) and (2) both give

$$\begin{bmatrix} -7 & 15 & 7 \\ 7 & -5 & 7 \end{bmatrix}$$

same result

Verification :

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \\ -3 & 5 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 0 & 4 \\ 4 & 0 & 8 \end{bmatrix} + \begin{bmatrix} -9 & 15 & 3 \\ 3 & -5 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2-9 & 0+15 & 4+3 \\ 4+3 & 0-5 & 8-1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2-9 & 0+15 & 4+3 \\ 4+3 & 0-5 & 8-1 \end{bmatrix}$$

$$AB = \begin{bmatrix} -7 & 15 & 7 \\ 7 & -5 & 7 \end{bmatrix}$$

$$AB = \begin{bmatrix} -7 & 15 & 7 \\ 7 & -5 & 7 \end{bmatrix}$$

Hence equal

Example : 12 (Trace)

$$B = \begin{bmatrix} -1 & 2 & 7 & 0 \\ 3 & 5 & -8 & 4 \\ 1 & 2 & 7 & -3 \\ 4 & -2 & 1 & 0 \end{bmatrix}$$

$$\text{tr}(B) = (-1) + (5) + (7) + (0)$$

$$\text{tr}(B) = 11$$

Exercise Set - 1.3

Question : 1

A	B	C	D	E
(4 × 5)	(4 × 5)	(5 × 2)	(4 × 2)	(5 × 4)

In below given part, Determine whether given matrix expression is defined. For those that are define given size of them.

a-

BA

- B is (4×5) , A is (4×5)
- Column of B (5) do not match row of A (4)
- So result is undefined.

b-

AB^T

- A is (4×5) , B^T is (5×4)
- Column of A is (5) and match rows of B^T (5)
- Result: Define
- Size: 4×4

c-

$AC + D$

- AC is (4×2) , D is (4×2)
- Since both are (4×2) so they can be added.
- Result: Define
- Size: (4×2)

d-

$E(AC)$

- E is (5×4) , AC is (4×2)
- No of cols of E (4) match rows of AC (4)
- Result: Define
- Size: (5×2)

$$A - 3E^T$$

e-

→ A is (4×5) , E^T is (4×5)

→ Both are of (4×5) , so can be subtracted.

→ Result: Define.

→ Size: (4×5)

f-

$$E(5B + A)$$

→ E is (5×4) , $(5B + A)$ are of (4×5)

→ Cols of E (4) matches rows of (4×5) $(5B + A)$

→ Result: Define

→ Size: (5×5)

Question: 2

Determine whether below expressions are defined
if yes give size of resulting matrix.

a-

$$CD^T$$

→ C is (5×2) , D^T is (2×4)

→ Cols of C (2) matches rows of D^T (2)

→ Result: Defined

→ Size: (5×4)

b-

$$DC$$

→ D is (4×2) , C is (5×2) , doesn't

→ Cols of D (2) matches rows of C (5)

→ Result: Undefined

c-

BC - 3D

- BC is (4×2) , 3D is (4×2)
- Both are of (4×2) so can be subtracted
- Result: Defined
- Size: (4×2)

d-

$D^T(BE)$

- D^T is (2×4) , and BE is (4×4)
- Cols of D^T (4) matches rows of BE (4)
- Result: Defined
- Size: (2×4)

e-

$B^T D + ED$

- $B^T D$ is (5×2) and ED is (5×2)
- Both are of same size so can be added
- Result: Defined
- Size: (5×2)

f-

$BA^T + D$

- BA^T is (4×4) and D is (4×2)
- BA^T (4×4) can't be added to (4×2)
- Result: Not Defined

Question: 3-b

Use following matrices to compute indicated expression if it is defined.

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}, \quad E = \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

Question: 3

a-

$D + E$

$$D + E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$D + E = \begin{bmatrix} 7 & 6 & 5 \\ -2 & 1 & 3 \\ 7 & 3 & 7 \end{bmatrix} \quad \text{Ans}$$

b-

$D - E$

$$D - E = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$D - E = \begin{bmatrix} -5 & 4 & -1 \\ 0 & -1 & -1 \\ -1 & 1 & 1 \end{bmatrix} \quad \text{Ans}$$

c-

5A

multiply matrix A by 5

d-

 $-7C$

multiply every entry of
matrix C by -7

e-

 $2B - C$

$\rightarrow B$ is (2×2) matrix

$\rightarrow C$ is (2×3) matrix

$\rightarrow$ so they can't be subtracted

$\rightarrow$ Result: Undefined

f-

 $4E - 2D$

$$= 4 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} - 2 \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & 4 & 12 \\ -4 & 4 & 8 \\ 16 & 4 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 10 & 4 \\ -2 & 0 & 2 \\ 6 & 4 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -6 & 8 \\ -2 & 4 & 6 \\ 10 & 0 & 4 \end{bmatrix} \quad \text{Ans}$$

g-

 $-3(D + 2E)$

$$-3 \left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + 2 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \right)$$

$$= -3 \left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 12 & 2 & 6 \\ -2 & 2 & 4 \\ 8 & 2 & 6 \end{bmatrix} \right)$$

$$= -3 \begin{bmatrix} 13 & 7 & 8 \\ -3 & 2 & 5 \\ 11 & 4 & 10 \end{bmatrix} \Rightarrow \begin{bmatrix} -39 & -21 & -24 \\ 9 & -6 & -15 \\ -33 & -12 & -30 \end{bmatrix} \text{ Ans}$$

h-

$A-A$

any matrix subtracted from itself
result in zero matrix of same size.

i-

$\text{tr}(D)$

Trace = sum of main diagonal entries.

$$\text{tr}(D) = 1 + 0 + 4 = 5 \text{ Ans}$$

j-

$\text{tr}(D-3E)$

using trace properties: $\text{tr}(D) - 3\text{tr}(E)$

$$\text{tr}(D) = 1 + 0 + 4 = 5$$

$$3\text{tr}(E) = 3(6 + 1 + 3) = 30$$

So,

$$\text{tr}(D-3E) = \text{tr}(D) - 3\text{tr}(E)$$

$$\text{tr}(D-3E) = 5 - 30$$

$$\text{tr}(D-3E) = 25 \text{ Ans}$$

k-

$$4 \text{tr}(7B)$$

$$\text{tr}(B) = 4 + 2 = 6$$

Using Trace Property

$$\text{tr}(kB) = k \text{tr}(B)$$

$$\text{tr}(7B) = 7 \text{tr}(B)$$

$$\text{tr}(7B) = 7(6)$$

$$\text{tr}(7B) = 42$$

$$4 \text{tr}(7B) = 4 \times 42$$

$$4 \text{tr}(7B) = 168 \text{ Ans}$$

L-

$$\text{tr}(A)$$

Not defined. Trace is only defined for square matrix.

Question: 4

a-

$$2A^T + C$$

$$= 2 \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}^t + \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 2 & 4 \\ 3 & 5 & 7 \end{bmatrix} \text{ Ans}$$

b-

$$D^T - E^t$$

$$= \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}^t - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}^t$$

$$= \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5 & 0 & -1 \\ 4 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix} \text{ Ans}$$

c-

Result will be same as part (b) according to property

$$\therefore (A+B)^T = A^T + B^T$$

$$= \begin{bmatrix} -5 & 0 & -1 \\ 4 & -1 & 1 \\ -1 & -1 & 1 \end{bmatrix} \text{ Ans}$$

d-

$$B^t + 5C^t$$

- B^t is (2×2) , C^t is (3×2)
- Both have different size, so can't be subtracted.
- Result: Undefined.

e-

$$\frac{1}{2} C^T - \frac{1}{4} A$$

→ C^T is (3×2) , A is (3×2)

→ So can be subtracted.

$$= \frac{1}{2} \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{3}{2} \\ 2 & \frac{1}{2} \\ 1 & \frac{5}{2} \end{bmatrix} - \begin{bmatrix} \frac{3}{4} & 0 \\ -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}$$

$$= \begin{bmatrix} -0.25 & 1.5 \\ 2.25 & 0 \\ 0.75 & 2.25 \end{bmatrix} \text{ Ans}$$

f-

$$B - B^T$$

$$= \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ Ans}$$

$$2E^T - 3D^T$$

$$= 2 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} - 3 \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix} - 3 \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -2 & 8 \\ 2 & 2 & 2 \\ 6 & 4 & 6 \end{bmatrix} - \begin{bmatrix} 3 & -3 & 9 \\ 15 & 0 & 6 \\ 6 & 3 & 12 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 1 & -1 \\ -13 & 2 & -4 \\ 0 & 1 & -6 \end{bmatrix} \text{ Ans}$$

h-
 $(2E^t - 3D^T)^T$

$$(2E^t - 3D^T)^t = \begin{bmatrix} 9 & 1 & -1 \\ -13 & 2 & -4 \\ 0 & 1 & -6 \end{bmatrix}^t$$

$$(2E^t - 3D^t)^t = \begin{bmatrix} 9 & -13 & 0 \\ 1 & 2 & 1 \\ -1 & -4 & -6 \end{bmatrix}$$

Ans

j-

$C(BA)$

→ C is (2×3)

→ BA is (2×2)

→ cols of C (3) doesn't match
rows of BA (2)

→ Result: Undefined

k-

$t\delta(DE^T)$

$$(DE^t) = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}^T$$

$$(DE^t) = \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix}$$

i-
 $(CD)E$

$$CD = \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}$$

$$CD = \begin{bmatrix} 1-4+6 & 5+0+4 & 2+4+8 \\ 3-1+15 & 15+0+10 & 6+1+20 \end{bmatrix}$$

$$CD = \begin{bmatrix} 3 & 9 & 14 \\ 17 & 25 & 27 \end{bmatrix}$$

$$(CD)E = \begin{bmatrix} 3 & 9 & 14 \\ 17 & 25 & 27 \end{bmatrix} \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}$$

$$(CD)E = \begin{bmatrix} 18-9+56 & 3+9+4 & 9+18+42 \\ 102-25+108 & 17+25+27 & 51+50+81 \end{bmatrix}$$

$$(CD)E = \begin{bmatrix} 65 & 26 & 69 \\ 185 & 69 & 182 \end{bmatrix} \text{ Ans}$$

$$(DE)^T = \begin{bmatrix} 6+5+6 & -1+5+4 & 4+5+6 \\ -6+0+3 & 1+0+2 & -4+0+3 \\ 18+2+12 & -3+2+8 & 12+2+12 \end{bmatrix} = \begin{bmatrix} 17 & 8 & 15 \\ -3 & 3 & -1 \\ 32 & 7 & 26 \end{bmatrix}$$

$$\text{tr}(DE^T) = 17+3+26$$

$$\text{tr}(DE^T) = 46 \text{ Ans}$$

L

$$\text{tr}(BC)$$

→ B is (2×2) , C is (2×3)

→ BC is (2×3) Not square matrix

→ Result: Undefined

Question: 5

a-

AB

→ A is (3×2) , B is (2×2)

→ cols of A match rows of B (2)

→ So Defined

→ Simply apply matrix multiplication

b-

BA

→ B is (2×2) , A is (3×2)

→ cols of B (2) doesn't match rows of A (3)

→ Result: Undefined

c-

$(3E)D$

→ E is (3×3) , D is (3×3)

→ So Defined

→ To get answer we can use matrix multiplication.

d-

$(AB)C$

→ A is (3×2) , B is (2×2)

→ AB is (3×2) , C is (2×3)

→ cols of AB (2) match rows of C (2)

→ Defined, to get answer use matrix multiplication.

i-

$$\text{tr}(DD^T)$$

$$= \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}^T \Rightarrow \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+25+4 & -1+0+2 & 3+10+8 \\ -1+0+2 & 1+0+1 & -3+0+4 \\ 3+10+8 & -3+0+4 & 9+4+16 \end{bmatrix} \Rightarrow \begin{bmatrix} 30 & 1 & 21 \\ 1 & 2 & 1 \\ 21 & 1 & 29 \end{bmatrix}$$

$$\text{tr}(DD^T) = 30+2+29$$

$$\text{tr}(DD^T) = 61 \text{ Ans}$$

j-

$$\text{tr}(4E^t - D)$$

$$= \left(4 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \right) \Rightarrow \left(4 \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 24 & -6 & 16 \\ 4 & 4 & 4 \\ 12 & 8 & 12 \end{bmatrix} - \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 23 & -9 & 14 \\ 5 & 4 & 3 \\ 9 & 6 & 8 \end{bmatrix}$$

$$\text{tr}(4E^t - D) = 23+4+8$$

$$\text{tr}(4E^t - D) = 35 \text{ Ans}$$

e-
 $A(BU)$

- A is (3×2) , BU is (2×3)
- cols of A (2) matches rows of BU (2)
- Defined, to get answer use matrix x-plication.

f-
 CC^T

- C is (2×3) , C^T is (3×2)
- cols of C (3) matches rows of C^T (3)
- Defined, to get ans. use matrix x-plication.

g-
 $(DA)^T$

$$= \left(\begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \right)^t$$

$$= \left(\begin{bmatrix} 3-5+2 & 0+10+2 \\ -3-0+1 & -0+0+1 \\ 9-2+4 & 0+4+4 \end{bmatrix} \right)^t = \left(\begin{bmatrix} 0 & 12 \\ -2 & 1 \\ 11 & 8 \end{bmatrix} \right)^t$$

$$= \begin{bmatrix} 0 & -2 & 11 \\ 12 & 1 & 8 \end{bmatrix} \text{ Ans.}$$

h-
 $(C^T B) A^T$

$$= \left(\begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}^t \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} \right) \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}^t$$

$$= \left(\begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix} \right) \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 4+0 & -1+6 \\ 16+0 & -4+2 \\ 8+0 & -2+10 \end{bmatrix} \right) \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 5 \\ 16 & -2 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 12+0 & -4+10 & 4+5 \\ 48-0 & -16-4 & 16-2 \\ 24+0 & -8+16 & 8+8 \end{bmatrix} = \begin{bmatrix} 12 & 6 & 9 \\ 48 & -20 & 14 \\ 24 & 8 & 16 \end{bmatrix} \text{ Ans}$$

K-

$$\text{tr}(C^T A^T + 2E^T)$$

$$= \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}^T \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}^T + 2 \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix}^T$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 1 \end{bmatrix} + 2 \begin{bmatrix} 6 & -1 & 4 \\ 1 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3+0 & -1+6 & 1+3 \\ 12+0 & -4+2 & 4+1 \\ 6+0 & -2+10 & 2+5 \end{bmatrix} + \begin{bmatrix} 12 & -2 & 8 \\ 2 & 2 & 2 \\ 6 & 4 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 5 & 4 \\ 12 & -2 & 5 \\ 6 & 8 & 7 \end{bmatrix} + \begin{bmatrix} 12 & -2 & 8 \\ 2 & 2 & 2 \\ 6 & 4 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & 3 & 12 \\ 14 & 0 & 7 \\ 12 & 12 & 13 \end{bmatrix}$$

$$\text{tr}(C^T A^T + 2E^T) = 15 + 0 + 13$$

$$\text{tr}(C^T A^T + 2E^T) = 28 \text{ Ans}$$

$$\text{tr}((EC^T)^T A)$$

$$= \left(\begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix}^T \right)^T \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 1 \\ 2 & 5 \end{bmatrix} \right)^T \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6+4+6 & 18+1+15 \\ -1+4+4 & -3+1+10 \\ 4+4+6 & 12+1+15 \end{bmatrix}^T \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 16 & 29 \\ 7 & 8 \\ 14 & 28 \end{bmatrix}^T \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 16 & 7 & 14 \\ 29 & 8 & 28 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 48-9+14 & 0+14+14 \\ 87-8+28 & 0+16+28 \end{bmatrix} \Rightarrow \begin{bmatrix} 55 & 28 \\ 107 & 44 \end{bmatrix}$$

$$= \text{tr}((EC^T)^T A) = 55 + 44$$

$$= \text{tr}((EC^T)^T A) = 99 \text{ Ans}$$

Question: 06

a-

$$(2D^T - E)A$$

$$\left(2 \begin{bmatrix} 1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \right) \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \left(2 \begin{bmatrix} 1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \right) \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \left(\begin{bmatrix} 2 & -2 & 6 \\ 10 & 0 & 4 \\ 4 & 2 & 8 \end{bmatrix} - \begin{bmatrix} 6 & 1 & 3 \\ -1 & 1 & 2 \\ 4 & 1 & 3 \end{bmatrix} \right) \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-6 & -2-1 & 6-3 \\ 10-1 & 0-1 & 4-2 \\ 4-4 & 2-1 & 8-3 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & -3 & 3 \\ 11 & -1 & 2 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -12+3+3 & 0-6+3 \\ 33-1+2 & 0-2+2 \\ 3-1+5 & 0+2+5 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & -3 \\ 36 & 0 \\ 4 & 7 \end{bmatrix} \text{ Ans}$$

b-

$$(4B)C + 2B$$

→ B is (2×2) , C is (2×3)

→ BC will be (2×3)

→ B is (2×2)

→ since BC and B is of different size

→ we can't add them so undefined.

c-

$$(-AC)^T + 5D^T$$

$$= \left(\begin{bmatrix} -3 & 0 \\ 1 & -2 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 2 \\ 3 & 1 & 5 \end{bmatrix} \right)^t + 5 \begin{bmatrix} +1 & 5 & 2 \\ -1 & 0 & 1 \\ 3 & 2 & 4 \end{bmatrix}^t$$

$$= \begin{bmatrix} -3+0 & -12+0 & -6+0 \\ 1-6 & 4-2 & 2-10 \\ -1-3 & -4-1 & -2-5 \end{bmatrix}^t + 5 \begin{bmatrix} +1 & -1 & 3 \\ 5 & 0 & 2 \\ 2 & 1 & 4 \end{bmatrix}^t$$

$$= \begin{bmatrix} -3 & -12 & -6 \\ -5 & 2 & -8 \\ -4 & -5 & -7 \end{bmatrix}^t + \begin{bmatrix} +5 & -5 & 15 \\ 25 & 0 & 10 \\ 10 & 5 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & -5 & -4 \\ -12 & 2 & -5 \\ -6 & -8 & -7 \end{bmatrix} + \begin{bmatrix} +5 & -5 & 15 \\ 25 & 0 & 10 \\ 10 & 5 & 20 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -10 & 11 \\ 13 & 2 & 5 \\ 4 & -3 & 13 \end{bmatrix} \quad \text{Ans}$$

d-

$$(BA^T - 2C)^T$$

Follow steps.

e-

$$B^T(C^T - A^T A)$$

↳ same procedure

f

$$D^T E^T - (ED)^T$$

↳ zero matrix

$$\therefore (AB)^{E^T T} = B^T A^T$$

Question: 7-8-

Use following matrices and either row operation or column method, as appropriate Find indicated row or column -

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

Question: 7

a-

First row of AB

$$= \begin{bmatrix} 3 & -2 & 7 \end{bmatrix} \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 18-0+49 & -6-2+49 & 12-6+35 \end{bmatrix}$$

$$= \begin{bmatrix} 67 & 41 & 41 \end{bmatrix}$$

↳ Ans

c-

Second Column of AB

$$= \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} -6-2+49 \\ -12+5+28 \\ 0+4+63 \end{bmatrix} = \begin{bmatrix} 41 \\ 21 \\ 67 \end{bmatrix}$$

↳ Ans

b-

Third row of AB

→ Third row of A

→ Full matrix B

d-

First column of AB

→ Full matrix A

→ First column of B

e-

Third row of AA

$$= \begin{bmatrix} 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 0+24+0 & 0+20+36 & 0+16+81 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & 56 & 97 \end{bmatrix} \text{ Ans}$$

f-

Third col. of AA

→ Full matrix A

→ Third col of A

Question : 8

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & -2 & 4 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

a-

The First Column
of AB

→ Full matrix A

→ First col of B

b-

Third column of BB

→ Full matrix B

→ Third col of B

c-

Second row of BB

→ second row of BB

→ Full matrix B

d-

First col of AA

→ Full matrix A

→ First col of A

e-

Third col. of AB

→ Full matrix A
 → Third col of B

f-

First row of BA

→ First row of B
 → Full matrix A

Question: 9

$$A = \begin{bmatrix} 3 & -2 & 7 \\ 6 & 5 & 4 \\ 0 & 4 & 9 \end{bmatrix}, B = \begin{bmatrix} 6 & -2 & 6 \\ 0 & 1 & 3 \\ 7 & 7 & 5 \end{bmatrix}$$

a-

Express each col vector of AA as a linear combinations of col vector of A.

1st col of AA:-

$$= 3 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 6 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

2nd col of AA:-

$$= -2 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 4 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

3rd col of AA.

$$= 7 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 9 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

b-

Express each col vector of BB as a linear comb. of col of B.

1st of BB:

$$= 6 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 0 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} + 7 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

2nd col of BB:

$$= -2 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 1 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} + 7 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

3rd col of BB

$$= 4 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} + 5 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

Question: 10-

same matrix from question 9

a-

Each col vector of AB as a linear comb. of col vector of A

1st col of AB

$$= 8 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 7 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

2nd col of AB

$$= -2 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 7 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

3rd col of AB :-

$$= 4 \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} -2 \\ 5 \\ 4 \end{bmatrix} + 5 \begin{bmatrix} 7 \\ 4 \\ 9 \end{bmatrix}$$

b-

Each col vector of BA as linear comb of col vectors of B.

1st col of BA

$$= 3 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 6 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} - 0 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

2nd col of BA

$$= -2 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 5 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} + 4 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

3rd col of BA

$$= 7 \begin{bmatrix} 6 \\ 0 \\ 7 \end{bmatrix} + 4 \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix} + 9 \begin{bmatrix} 4 \\ 3 \\ 5 \end{bmatrix}$$

Note:- Each col vector of BA as a linear combination of vector of B (invalid question coz order matter).

Question : 11-12.

Find matrices A, x and b that express given linear system as a single expression $AX=b$ write out this equation.

11-(a)

$$\begin{aligned} 2x_1 - 3x_2 + 5x_3 &= 7 \\ 9x_1 - x_2 + x_3 &= -1 \\ x_1 + 5x_2 + 4x_3 &= 0 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 9 & -1 & 1 \\ 1 & 5 & 4 \end{bmatrix}, b = \begin{bmatrix} 7 \\ -1 \\ 0 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 & 5 \\ 9 & -1 & 1 \\ 1 & 5 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ 0 \end{bmatrix} \quad \text{Ans.}$$

Q:12: Same as question 11 and part b of Q:11 too

Question: 13-14

13-(b)

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 5 & -3 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ -9 \end{bmatrix}$$

$$\begin{aligned} x + y + z &= 2 \\ 2x + 3y &= 2 \\ 5x - 3y - 6z &= -9 \end{aligned}$$

Q:14 and Q:13 (a) are same like above question.

Question: 15-16-

Find all values of k if any that satisfy the equation

15-

$$\begin{bmatrix} k & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

$\rightarrow (1 \times 3), (3 \times 3), (3 \times 1)$

$$\begin{bmatrix} k+1+0 & k+0+2 & 0+2-3 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} k+1 & k+2 & -1 \end{bmatrix} \begin{bmatrix} k \\ 1 \\ 1 \end{bmatrix} = 0$$

$$k(k+1) + 1(k+2) + (-1)(1) = 0$$

$$k^2 + k + k + 2 - 1 = 0$$

$$k^2 + 2k + 1 = 0$$

$$(k+1)^2 = 0$$

Taking sq. r b/s

$$\boxed{k = -1}$$

16-

$$\begin{bmatrix} 2 & 2 & k \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ k \end{bmatrix} = 0$$

$$\begin{bmatrix} 2+2+0 & 4+0+3k & 0+6+k \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ k \end{bmatrix} = 0$$

$$\begin{bmatrix} 6 & 3k+4 & k+6 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ k \end{bmatrix} = 0$$

$$(6)(2) + 2(3k+4) + k(k+6) = 0$$

$$12 + 6k + 8 + k^2 + 6k = 0$$

$$k^2 + 12k + 20 = 0$$

$$k^2 + 10k + 2k + 20 = 0$$

$$k(k+10) + 2(k+10) = 0$$

$$(k+2)(k+10) = 0$$

$$\boxed{k = -2}, \boxed{k = -10}$$

Question : 17-20

Use column row expansion of AB to express this as sum of matrices.

17-

$$A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 2 \\ -2 & 3 & 1 \end{bmatrix}$$

$$\therefore AB = c_1 r_1 + c_2 r_2$$

$$AB = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} + \begin{bmatrix} -3 \\ -1 \end{bmatrix} \begin{bmatrix} -2 & 3 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 4 & 8 \\ 0 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 6 & -9 & -3 \\ 2 & -3 & -1 \end{bmatrix} \quad \text{Ans}$$

18-

Same as Q. 17

$$A = \begin{bmatrix} 0 & -2 \\ 4 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 & 1 \\ -3 & 0 & 2 \end{bmatrix}$$

19-

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$\therefore AB = c_1 r_1 + c_2 r_2 + c_3 r_3$$

$$AB = \begin{bmatrix} 1 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} + \begin{bmatrix} 3 \\ 6 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix} + \begin{bmatrix} 6 & 8 \\ 15 & 20 \end{bmatrix} + \begin{bmatrix} 15 & 18 \\ 30 & 36 \end{bmatrix} \quad \text{Ans}$$

20-

Same as Q: 19

$$A = \begin{bmatrix} 0 & 4 & 2 \\ 1 & -2 & 5 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 \\ 4 & 0 \\ 1 & -1 \end{bmatrix}$$

Question: 23-24

Solve the matrix for a, b, c and d.

2 -

$$\begin{bmatrix} a & 3 \\ -1 & a+b \end{bmatrix} = \begin{bmatrix} 4 & d-2c \\ d+2c & -2 \end{bmatrix}$$

$$\begin{aligned} \boxed{a=4} &\rightarrow (1) & 3 &= d-2c \rightarrow (2) \\ -1 &= d+2c \rightarrow (3) & a+b &= -2 \rightarrow (4) \end{aligned}$$

put eq (1) in eq (4)

$$a+b = -2$$

$$4+b = -2$$

$$\boxed{b = -6}$$

add eq (2) and (3)

$$3 = d-2c$$

$$-1 = d+2c$$

$$\boxed{d = 1}$$

put d = 1 in eq (2)

$$3 = d-2c$$

$$3 = 1-2c \Rightarrow \boxed{c = -1}$$

2 -

$$\begin{bmatrix} a-b & b+a \\ 3d+c & 2d-c \end{bmatrix} = \begin{bmatrix} 8 & 1 \\ 7 & 6 \end{bmatrix}$$

$$a-b = 8 \rightarrow (1)$$

$$a+b = 1 \rightarrow (2)$$

$$3d+c = 7 \rightarrow (3)$$

$$2d-c = 6 \rightarrow (4)$$

add eqn (1) and eqn (2)

$$+ a - b = +8$$

$$+ a + b = +1$$

$$2a = 9$$

$$a = \frac{9}{2}$$

put $a = \frac{9}{2}$ in eqn (2)

$$a + b = 1$$

$$\frac{9}{2} + b = 1$$

$$b = 1 - \frac{9}{2}$$

$$b = -\frac{7}{2}$$

add eqn (3) and eqn (4)

$$3d + c = 7$$

$$2d - c = 6$$

$$5d = 13$$

$$d = \frac{13}{5}$$

put $d = \frac{13}{5}$ in eqn (3)

$$3d + c = 7$$

$$3\left(\frac{13}{5}\right) + c = 7$$

$$\frac{39}{5} + c = 7$$

$$c = 7 - \frac{39}{5}$$

$$c = -\frac{4}{5}$$

Question: 25

a-

Ans:- If A has row of zeros, then AB will also have a row of zeros in same position.

b-

Ans:- If B has col of zero, then AB will also have a column of zeroes in same position.

Question: 26-

In each part find 6×6 matrix $[a_{ij}]$ that satisfies stated condition. Make your ans as a general as possible by using letters rather than specific numbers for non zero entries.

a-

$$a_{ij} = 0 \quad \text{if } i \neq j$$

So.

a	0	0	0	0	0
0	b	0	0	0	0
0	0	c	0	0	0
0	0	0	d	0	0
0	0	0	0	e	0
0	0	0	0	0	f

→ Ans.

a_{11}	a_{12}	a_{13}	a_{14}	a_{15}	a_{16}
a_{21}	a_{22}	a_{23}	a_{24}	a_{25}	a_{26}
a_{31}	a_{32}	a_{33}	a_{34}	a_{35}	a_{36}
a_{41}	a_{42}	a_{43}	a_{44}	a_{45}	a_{46}
a_{51}	a_{52}	a_{53}	a_{54}	a_{55}	a_{56}
a_{61}	a_{62}	a_{63}	a_{64}	a_{65}	a_{66}

b-

$$a_{ij} = 0 \text{ if } i > j$$

a	b	c	d	e	f
0	g	h	i	j	k
0	0	l	m	n	o
0	0	0	p	q	r
0	0	0	0	s	t
0	0	0	0	0	u

Ans.

c-

$$a_{ij} = 0 \text{ if } i < j$$

a	0	0	0	0	0
b	c	0	0	0	0
d	e	f	0	0	0
g	h	i	j	0	0
k	l	m	n	0	0
p	q	r	s	t	u

Ans.

d-

$$a_{ij} = 0 \text{ if } |i-j| > 1$$

$$\begin{bmatrix} a & b & 0 & 0 & 0 & 0 \\ c & d & e & 0 & 0 & 0 \\ 0 & f & g & h & 0 & 0 \\ 0 & 0 & i & j & k & 0 \\ 0 & 0 & 0 & l & m & n \\ 0 & 0 & 0 & 0 & 0 & p \end{bmatrix}$$

$$|1-1| = 0$$

$$|1-2| = 1$$

$$|1-3| = 2$$

Question: 32-

Find 4×4 matrix $A = [a_{ij}]$ whose entries satisfy stated condition.

a-

$$a_{ij} = i + j$$

$$\begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 \end{bmatrix} \quad \text{Ans}$$

b-

$$a_{ij} = i^{j-1}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{bmatrix} \quad \text{Ans}$$

C-

$$a_{ij} = \begin{cases} 1 & \text{if } |i-j| > 1 \\ -1 & \text{if } |i-j| \leq 1 \end{cases}$$

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 \\ 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{bmatrix} \quad \text{Ans.}$$

Question: 33

Suppose that Type 1 item cost \$1 each, type II item cost \$2 each and Type III items cost \$3 each. Also suppose that the accompanying table describes number of items of each type purchased during first four months of year.

	Type 1	Type 2	Type 3
Jan.	3	4	3
Feb.	5	6	0
Mar.	2	9	4
Apr.	1	1	7

Type	cost
1	\$1
2	\$2
3	\$3

what info is represented by following product?

$$\begin{bmatrix} 3 & 4 & 3 \\ 5 & 6 & 0 \\ 2 & 9 & 4 \\ 1 & 1 & 7 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 3+8+9 \\ 5+12+0 \\ 2+18+12 \\ 1+2+21 \end{bmatrix} \Rightarrow \begin{bmatrix} 20 \\ 17 \\ 32 \\ 24 \end{bmatrix}$$

→ so this vector represent total cost of each item purchased in each months.

Question: 34

The below tables shows record of May and June unit sales for clothing store. Let M denote 4×3 matrix of May sales and J 4×3 matrix of June sales

May sales				June sales			
	Small	Medium	Large		Small	Med	Large
Shirts	45	60	75	Shirts	30	33	40
Jeans	30	30	40	Jeans	21	23	25
Suits	12	65	45	Suits	9	12	11
Raincoats	15	40	35	Raincoats	8	10	9

a-

What $M+J$ represent?

$$\begin{bmatrix} 45 & 60 & 75 \\ 30 & 30 & 40 \\ 12 & 65 & 45 \\ 15 & 40 & 35 \end{bmatrix} + \begin{bmatrix} 30 & 33 & 40 \\ 21 & 23 & 25 \\ 9 & 12 & 11 \\ 8 & 10 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 75 & 93 & 115 \\ 51 & 53 & 65 \\ 21 & 77 & 56 \\ 23 & 50 & 44 \end{bmatrix} \rightarrow \text{So } M+J \text{ represent total sales of each item and size during May and June together.}$$

b-

$$\begin{bmatrix} 45 & 60 & 75 \\ 30 & 30 & 40 \\ 12 & 65 & 45 \\ 15 & 40 & 35 \end{bmatrix} - \begin{bmatrix} 30 & 33 & 40 \\ 21 & 23 & 25 \\ 9 & 12 & 11 \\ 8 & 10 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 15 & 27 & 35 \\ 9 & 7 & 15 \\ 3 & 53 & 34 \\ 7 & 30 & 26 \end{bmatrix} \rightarrow \text{So } M-J \text{ represents difference in sales b/w May and June For each item and size.}$$

c-

Find a column vector x for which Mx provides a List of numbers of shirts, jeans, suits, and Raincoats sold in May.

So x will be $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

$$= \begin{bmatrix} 45 & 60 & 75 \\ 30 & 30 & 40 \\ 12 & 65 & 45 \\ 15 & 40 & 35 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 45+60+75 \\ 30+30+40 \\ 12+65+45 \\ 15+40+35 \end{bmatrix}$$

$$= \begin{bmatrix} 180 \\ 100 \\ 122 \\ 90 \end{bmatrix} \rightarrow \text{No of Shirts, Jeans, Suits and Raincoats sold in May.}$$

d-

Find row vector y for which yM provides a List of number of small, medium and large item sold in May.

So y will be $\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 45 & 60 & 75 \\ 30 & 30 & 40 \\ 12 & 65 & 45 \\ 15 & 40 & 35 \end{bmatrix} \Rightarrow \begin{bmatrix} 45+30+12+15 & 60+30+65+40 & 75+40+45+35 \end{bmatrix}$$

$$= \begin{bmatrix} 102 & 195 & 195 \end{bmatrix} \rightarrow \text{Total no. of items sold in May For each size.}$$

d-

using matrix x and y and find xMy .

as we have calculate xMy or Mx in previous part so we can use one of them

$$= \begin{bmatrix} 102 & 195 & 195 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 102 + 195 + 195 \end{bmatrix}$$

$$= \begin{bmatrix} 492 \end{bmatrix}$$

$\rightarrow$ Total no. of all clothing items sold in May across all sizes.